

# Alternatives and Safety Measures for 4-Liter Chemical Bottles

## Alternatives to 4-Liter Chemical Bottles

Sometimes chemical spills happen when researchers drop 4-liter bottles. Although ubiquitous in laboratories, these bottles can be heavy and awkward to handle. Incidents can be prevented by purchasing chemicals in:

**Bottles that contain volumes less than 4 liters**

Plastic-coated 4-L bottles (pictured below)



## Safe Practices for Handling and Storage

If alternatives are not feasible, follow these safe practices to minimize risks:

Do not lift or hold a 4-Liter bottle solely by its small handle; instead, grasp a bottle firmly with two hands around the center, or place one hand underneath and the other around the side

**Use a bottle carrier (pictured below) or secondary container with a cart to transport the bottle**

Lift safely using correct posture

**Place 4-L bottles on low shelves**

Store bottles in secondary containment segregated by chemical storage groups



## Spill Response Information

Call 725-9999 for cleanup assistance whenever:

**The spill poses an inhalation hazard**

You cannot clean up a spill safely yourself in 15 minutes

**The spill is not contained in a chemical fume hood or biosafety cabinet, or on a lab bench**

The spill may enter a sink or floor drain, or contaminate soil

**To request a spill kit (pictured below) and other hazardous waste supplies, visit [safetystore.stanford.edu](https://safetystore.stanford.edu).**

